

AR40215

Building Energy System and Auditing

Semester-VI, B.Arch

Building Energy Index & Auditing Energy Audit



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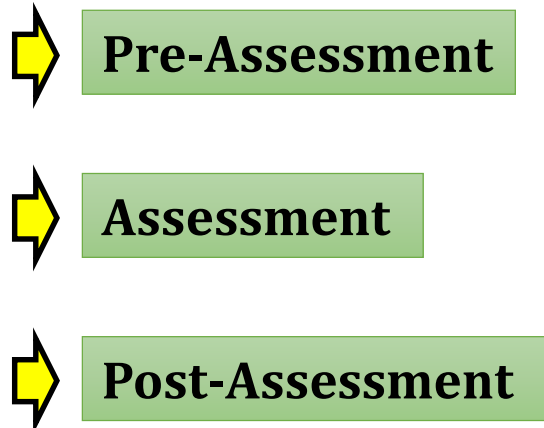
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Definition

The building energy audit is a study carried out to identify energy use in various services and sectors to identify opportunities for energy conservation. It is also a crucial first step in establishing an energy management program. The study should reveal to the owner, manager, or management team of the building the options available for reducing energy waste, the costs involved, and the benefits achievable from implementing those energy-conserving opportunities.

Stages of Energy Audit and Assessment



Pre-Assessment

Pre-Assessment: In this stage various actions are taken to set up the higher level of assessment. This stage will outline the need, scope and objective of the assessment process. The pre assessment step includes:

- Benchmarking and Screening to decide if the audit is necessary
- Preliminary data collection: Building information (size, age), Type of building, Building use, one year electricity bill.
- Formation of energy Efficiency Committee
- Set the performance target
- Determine the level of assessment to pursue

Assessment

The assessment section gives an overview of each level of building energy assessment. It also states the level of effort and prospective results can be expected out of the survey. This stage includes:

Data collection and analysis for the following three level of assessment

Level	Basic type	Purpose
Level-I Assessment	Preliminary Energy use and walk through analysis	• Get a base line for energy use. • Full understanding of how energy is being consumed and associated cost. • Forecast the future energy demand. • Set a preliminary target for energy reduction.
Level-II Assessment	Energy survey and analysis	• Execute a detail and physical energy audit programme. • Create a full energy conservation programme. • Evaluate all potential energy conserving programmes.
Level-III Assessment	Details analysis of capital investment measure	• To undertake significant capital projects to improve the energy efficiency based on the project identified in the previous level of assessment. • Provide the detailed and high level of confidence report to client to make decision to execute the selected programme.

Post-Assessment

Post Assessment: The post-assessment section provides information on how to implement and continue the ongoing energy saving based on the result of the building energy assessment. This stage includes:

- Measurement and verification of the result obtained in the assessment stage
- Implementation of energy saving projects
- Application of Operation and Maintenance guidelines
- Overall review

TYPES OF ENERGY AUDITS

Generally, there are four types of energy audits:

1. Walk-Through Audit

2. Utility Cost Analysis Audit

3. Standard Energy Audit

4. Detailed Energy Audit

1. Walk-Through Audit

This is shortest type energy audit. It consists of a short onsite visit for the inspection of the facility. Visual observation leads to some simple actions and solutions for immediate energy savings. The final outcome of the audit consists of :

- Repairing broken glass windows
- Lowering the pre-set temperatures of HVAC systems according to utility
- Adjusting the boiler-air fuel ratios

This is usually a maintenance procedure done periodically to improve the efficiency of energy systems.

2. Utility Cost Analysis Audit

This type of audit analyses the energy operating cost of the building and facility. The data obtained over a long period of time energy bills, peak demands, energy use patterns, weather effects are identified. This helps us to establish a relation between cost and utility. Usually this step includes:

- Checking utility services and corresponding monthly energy bills.
- Analysis of peak demands and major share of the power consumption cost.
- Checking the probable benefit from alternative fuels which are more cost effective than the prevailing ones.
- Possibilities of energy retrofit in the projects by analysing the utility data.

3. Standard Energy Audit

The standard audit provides a comprehensive analysis of the energy systems of the building and facility. In addition to earlier two types, the standard energy audit includes:

- The development of a baseline for the energy use of the building and facility
- The evaluation of the energy savings and the cost effectiveness of appropriately selected energy conservation measures.
- The step by step approach and simplified tools are used in the standard energy audit to develop baseline energy models and to predict the energy savings of energy conservation measures. Among these tools are the degree-day methods, and linear regression models.
- A simple payback analysis is generally performed to determine the cost-effectiveness of energy conservation measures.

4. Detailed Energy Audit

This is the most comprehensive type of energy audit. But it is a time-consuming method.

The detailed energy audits include the use of instruments to measure energy use for the whole building. In addition, sophisticated computer simulation programs are typically employed for detailed energy audits to evaluate and recommend energy retrofits for the facility.

They are often based on dynamic thermal performance of the building energy systems and usually require a high level of engineering expertise and training. In the detailed energy audit, more rigorous economical evaluation of the energy conservation measures is generally performed. Specifically, the cost-effectiveness of energy retrofits may be determined based on the life-cycle cost (LCC) analysis rather than the simple payback period analysis. LCC analysis takes into account a number of economic parameters such interest, inflation, and tax rates.